



solid fertiliser dispensing. Unlike traditional methods involving manual spreading or spraying, Krishi Dhanu directly dispenses solid fertiliser. It operates with two motors for grinding and dispersing the fertiliser and includes a moderator to control the application rate. The device features an IoT-enabled weighing scale that records and stores data on fertiliser usage and types. It also integrates with crop and soil monitoring applications, providing insights into fertiliser requirements and soil conditions.

“We have filed for three patents for Dhivara Mitra, covering its DC motor, design pattern, and feeding solution, with two already granted. These patents include the overall design and the feed dispensing system. Additionally, we have applied for a patent for Krishi Dhanu, which is expected to be approved within the next one to two months,” shares Minushri.

They claim to have procured the glass-fibre-based bodies of Dhivara Mitra and Krishi Dhanu from a renowned shipbuilding company in India, which ranks as the country’s second-best. While customising motors sourced from the market, their in-house capabilities extend to acquiring solar panels and other components. They fabricate the vessel’s structure and internal components in-house, including sensors, solar panels, and other IoT applications developed through their R&D efforts. However, for hardware devices, they rely on market availability.

Fifteen units of Krishi Dhanu have been sold: five to the Farmer Federation aided by the Odisha Livelihoods Mission in the Kalahandi district of Odisha, and the remaining ten units allocated to FPOs supported by SELCO Foundation.

Depending on the features, Dhivara Mitra boats are priced at



Minushri Madhumita, Co-Founder, ThinkRaw India

₹550,000 to ₹660,000 per unit. The ₹660,000 tier includes sensors for digital oxygen and pH level measurement, which are suitable for farmers with one-acre land. Some opt for the cheaper option due to sensor costs. Government subsidies cover 40-50% in Odisha and 30% from the National Fishery Development Board across India.

If someone cannot afford to buy it outright, they can rent it. “As mentioned earlier, five units of Dhivara Mitra have been sold to a farmer-producer company, and one unit has been rented out to a farmer at a monthly rate of ₹10,000,” says Minushri.

“The ten units of Krishi Dhanu sold to the Odisha Livelihoods Mission-supported Farmer Federation are rented out by various Farmer Producer Organisations (FPOs). Here is how it works: Let’s consider a self-help group with 14 members. One member is trained to operate the machine, and they collectively cover the labour charges and machine rental, amounting to around ₹1000 per day. This setup helps them efficiently manage

agricultural tasks despite labour shortages,” she adds.

The startup has an assembly unit in Paradeep, Odisha, and is seeking funds to expand its business. ThinkRaw employs 11 individuals and generates revenue by selling solar setups and providing rental services for market-ready versions of its products.

The startup follows two potential business approaches: The first involves directly engaging with large-scale farmers who buy in bulk, while the second focuses on collaborating with FPOs, which have member bases ranging from 300 to 2000. The first approach involves direct sales to farmers, while the second involves selling through FPOs, which receive a 10% commission on sales.

Societal perceptions of farming as a low-status occupation hinder efforts to convince farmers to adopt our products despite their affordability. “Our challenge lies in showcasing product benefits, altering entrenched views on farming, and recognising farmers’ progressive nature,” adds Minushri.

The startup plans to establish a presence in Canada within the next few months. Minushri shares, “With 30% of Canada’s economy stemming from fish and prawn farms, we see tremendous potential in this market. Furthermore, we are eyeing opportunities in other American countries and Asian nations like Indonesia and Bangladesh. Our goal is to become a truly global company.”

From 2016 to 2023, their revenue, team size, and brand recognition have doubled, surpassing all expectations. Despite starting with minimal investment, they have achieved remarkable growth, with revenues now exceeding ₹20 million. **EFY**

—Nitisha Dubey